آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 3

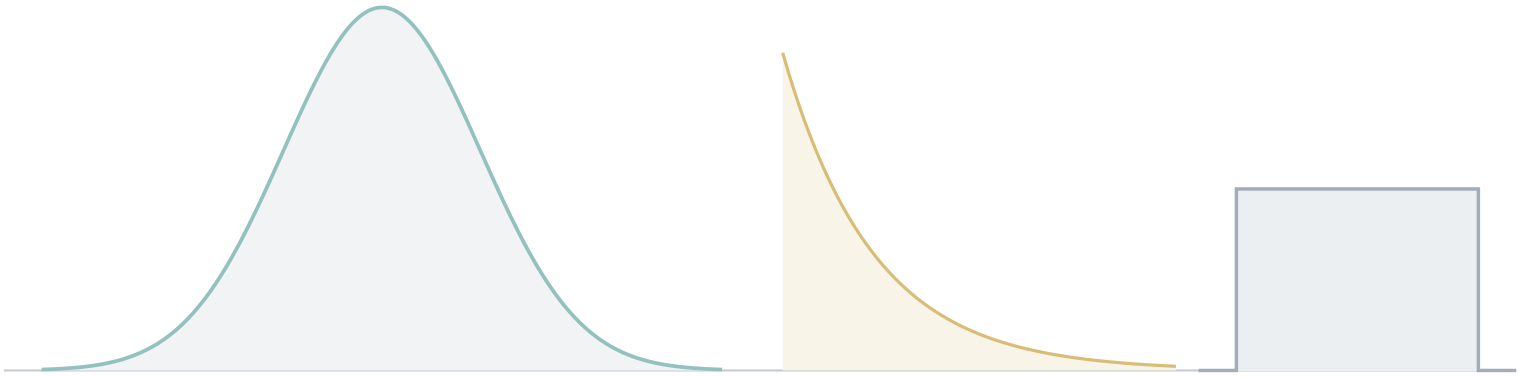
Important Probability Distributions

توزیع‌های مهم احتمال

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Learning Objectives

After completing this module, students will be able to:

1. Identify which probability distribution models a given engineering scenario
2. State the $\frac{\text{PMF}}{\text{PDF}}$, CDF, mean, and variance for all major distributions
3. Explain the physical interpretation and engineering context for each distribution
4. Determine when to use (and when NOT to use) each distribution
5. Analyze and compare probability distributions
6. Recognize distribution patterns from data characteristics

3.1 Introduction: Choosing the Right Distribution

The Engineering Challenge

When modeling a random phenomenon, the critical question is:

> "Which probability distribution best describes this uncertain quantity?"

Choosing the wrong distribution leads to incorrect predictions, improper designs, and system failures.

How to Choose

The distribution should match the **physical mechanism** generating randomness:

- Counting successes → Binomial
- Waiting for events → Exponential/Geometric
- Sum of many small effects → Gaussian
- Signal envelope in fading → Rayleigh

This module provides the tools to make this choice correctly.

3.2 Bernoulli Distribution

Physical Interpretation

Models a **single trial** with two possible outcomes: "success" (1) or "failure" (0).

Engineering Context

Any binary random outcome:

- One bit transmitted: error or correct
- One component tested: pass or fail
- One packet sent: delivered or lost
- One detection attempt: target present or absent

Mathematical Definition

$X \sim$ Bernoulli(p), where p = probability of success.

PMF

$$p_X(x) = \begin{cases} p & x = 1 \\ 1 - p = q & x = 0 \end{cases}$$

Compact form: $p_X(x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$

CDF

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - p & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

Mean

$$E[X] = p$$

Variance

$$\text{Var}(X) = p(1-p)$$

Maximum variance at $p = 0.5$ (maximum uncertainty).

When to Use

✔ Single binary trial with fixed probability
✔ Building block for more complex distributions
✔ Modeling $\frac{ON}{OFF}$ states, pass/fail outcomes

When NOT to Use

✘ Multiple trials (use Binomial)
✘ Outcome is not binary
✘ Probability changes between trials

3.3 Binomial Distribution

Physical Interpretation

Models the **number of successes** in n **independent** trials, each with the same success probability p.

Engineering Context

- Number of bit errors in n transmitted bits
- Number of defective components in a batch of n
- Number of successful packet transmissions out of n attempts
- Number of sensors that detect a signal (out of n sensors)

Mathematical Definition

$X \sim \text{Binomial}(n,\, p)$

PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \ldots, n$$

where $C(n,\, k) = \frac{n!}{k!(n-k)!}$

CDF

$$F_X(k) = \sum_{i=0}^{\lfloor k \rfloor} \binom{n}{i} p^i (1-p)^{n-i}$$

No closed-form expression — use tables or numerical summation.

Mean

$$E[X] = np$$

Variance

$$\text{Var}(X) = np(1-p)$$

When to Use

✔ Fixed number of trials n
✔ Each trial is independent
✔ Each trial has the same probability p
✔ Counting the number of "successes"

When NOT to Use

✘ Trials are not independent (correlated failures)
✘ Probability changes between trials
✘ n is not fixed (use Negative Binomial or Poisson)
✘ Very large n with small p (use Poisson approximation)

3.4 Geometric Distribution

Physical Interpretation

Models the **number of trials until the first success** (or equivalently, the number of failures before the first success).

Engineering Context

- Number of transmissions until a packet is successfully delivered
- Number of components tested until finding a defective one
- Number of login attempts until successful authentication
- Number of time slots until channel becomes available

Mathematical Definition

$X \sim \text{Geometric}(p)$
Convention: X = number of the trial on which first success occurs ($X \in \{1, 2, 3, \ldots\}$).

PMF

$$p_X(k) = (1-p)^{k-1}p, \quad k = 1, 2, 3, \ldots$$

($k-1$ failures, then one success)

CDF

$$F_X(k) = 1 - (1-p)^k, \quad k = 1, 2, 3, \ldots$$

Mean

$$E[X] = \frac{1}{p}$$

Variance

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Memoryless Property

The geometric distribution is the ONLY discrete distribution with the memoryless property:

$$P(X > m+n \mid X > m) = P(X > n)$$

Engineering meaning: If a system hasn't succeeded in m attempts, the probability of needing n more attempts is the same as starting fresh. "Past failures don't affect future probability."

When to Use

✔ Repeated independent trials until first success
✔ Each trial has the same probability
✔ Memoryless assumption is appropriate (e.g., independent retransmissions)

When NOT to Use

✘ Success probability changes over time (aging, learning)
✘ Trials are not independent
✘ Looking for the $r-$ th success (use Negative Binomial)

3.5 Negative Binomial Distribution

Physical Interpretation

Models the **number of trials until the** $r-$ th **success** (generalization of Geometric).

Engineering Context

- Number of transmissions until r packets are successfully delivered
- Number of components inspected until finding r defectives
- Number of time slots until r channels become available

Mathematical Definition

$X \sim \text{NegBin}(r,\, p)$ = number of trials until $r-$ th success.

PMF

$$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \ldots$$

(Choose which $r-1$ of the first $k-1$ trials were successes, the $k-$ th is the $r-$ th success.)

CDF

No simple closed form. Computed by summation or numerical methods.

Mean

$$E[X] = \frac{r}{p}$$

Variance

$$\text{Var}(X) = \frac{r(1-p)}{p^2}$$

Relationship to Geometric

When $r = 1$, the Negative Binomial reduces to the Geometric distribution.

When to Use

✔ Repeated independent trials until $r-$ th success
✔ Constant probability per trial
✔ Generalizing geometric to multiple required successes

When NOT to Use

✘ Only need first success (use Geometric — simpler)
✘ Trials not independent
✘ Fixed number of trials (use Binomial)

3.6 Poisson Distribution

Physical Interpretation

Models the **number of events occurring in a fixed interval** (time, space, area) when events occur independently at a constant average rate.

Engineering Context

- Number of photons hitting a detector in $1\mu s$
- Number of packets arriving at a router in 1 second
- Number of defects per cm^2 on a silicon wafer
- Number of cosmic ray events in a sensor per hour
- Number of interference spikes in a measurement window

Mathematical Definition

$X \sim \text{Poisson}(\lambda)$, where λ = average number of events per interval.

PMF

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

CDF

$$F_X(k) = e^{-\lambda} \sum_{i=0}^{\lfloor k \rfloor} \frac{\lambda^i}{i!}$$

Mean

$$E[X] = \lambda$$

Variance

$$\text{Var}(X) = \lambda$$

Unique property: Mean equals variance! This is a quick check — if sample mean ≈ sample variance, data may be Poisson.

Poisson as Binomial Approximation

When n is large and p is small, with $\lambda = np$:

$$\text{Binomial}(n, p) \approx \text{Poisson}(\lambda = np)$$

Rule of thumb: $n \geq 20$ and $p \leq 0.05$.

When to Use

✔ Counting events in a fixed interval
✔ Events occur independently
✔ Events occur at a constant average rate
✔ Rare events with many opportunities (large n, small p)
✔ Mean $\approx$ Variance in data

When NOT to Use

✘ Events are not independent (clustered arrivals)
✘ Rate varies with time (non-homogeneous process)
✘ Events are not rare (p not small) — use Binomial
✘ Mean $\neq$ Variance (look at overdispersion/underdispersion)

3.7 Uniform Distribution (Continuous)

Physical Interpretation

Models a quantity that is **equally likely** to take any value in an interval $[a, b]$. Represents **maximum uncertainty** within a known range.

Engineering Context

- Phase of a received signal (unknown, equally likely in $[0, 2\pi)$)
- Round-off error in an ADC (uniform in $\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]$ where Δ = quantization step)
- Arrival time within a time slot (unknown position)
- Random access protocol: choose a random backoff time

Mathematical Definition

$X \sim \text{Uniform}(a, b)$ or $X \sim U(a, b)$

PDF

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

CDF

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

Mean

$$E[X] = \frac{a+b}{2}$$

Variance

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

When to Use

✔ All values in a range are equally likely
✔ No information favors any particular value
✔ Quantization error modeling
✔ Phase of unmodulated carrier

When NOT to Use

✘ Values near the center are more likely (use Gaussian)
✘ Values are non-negative with exponential decay (use Exponential)
✘ Range is unbounded

3.8 Exponential Distribution

Physical Interpretation

Models the **time between events** in a Poisson process, or the **lifetime** of a memoryless component.

Engineering Context

- Time between packet arrivals
- Time until component failure (constant failure rate)
- Time between cosmic ray events in a detector
- Inter-arrival time of calls at a switch
- Duration of a fading dip in a wireless channel

Mathematical Definition

$X \sim$ Exponential(λ) where λ = rate parameter (events per unit time).
Alternative parameterization: $X \sim \text{Exp}(\beta)$ where $\beta = \frac{1}{\lambda}$ = mean.

PDF

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Or with mean β: $f_X(x) = \frac{1}{\beta} e^{-x/\beta}$

CDF

$$F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

Mean

$$E[X] = \frac{1}{\lambda} = \beta$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2} = \beta^2$$

Note: Standard deviation = mean for exponential distribution.

Memoryless Property

$$P(X > s + t \mid X > s) = P(X > t)$$

The exponential is the ONLY continuous distribution with this property.

Engineering meaning: A component that has been working for s hours is statistically "as good as new." Its remaining lifetime has the same distribution as a brand-new component.

Relationship to Poisson

If events arrive as a Poisson process with rate λ:

- Number of events in time $T \sim \text{Poisson}(\lambda T)$
- Time between events $\sim \text{Exponential}(\lambda)$

When to Use

✔ Modeling time between events in a Poisson process
✔ Component lifetime with constant failure rate
✔ Memoryless assumption is reasonable

When NOT to Use

✘ Failure rate increases with age (wear-out) — use Weibull or Gamma
✘ Failure rate decreases with age (burn-in) — use Weibull
✘ Data shows increasing hazard — NOT memoryless

3.9 Gaussian (Normal) Distribution

Physical Interpretation

Models quantities that result from the **sum of many small, independent random effects**. The most important distribution in engineering due to the Central Limit Theorem.

Engineering Context

- Thermal noise voltage (sum of many electron contributions)
- Measurement errors (sum of many small error sources)
- Manufacturing variations (many independent process variations)
- Aggregate interference in communications
- Any quantity well-modeled by CLT

Mathematical Definition

$X \sim N(\mu,\, \sigma^2)$ where μ = mean, σ^2 = variance.

PDF

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

CDF

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$$

No closed-form — computed numerically or via tables.

Mean

$$E[X] = \mu$$

Variance

$$\text{Var}(X) = \sigma^2$$

Standard Normal

$Z \sim N(0,\, 1)$: standardized form.

Any Normal can be standardized: $Z = \frac{X-\mu}{\sigma}$

Key Probability Rules (σ − rules)

Range	Probability
$\mu \pm 1\sigma$	68.27%
$\mu \pm 2\sigma$	95.45%

Range	Probability
$\mu \pm 3\sigma$	99.73%
$\mu \pm 4\sigma$	99.9937%

Properties

1. **Symmetric** about μ
2. **Linear transformation:** If $X \sim N(\mu,\, \sigma^2)$, then aX $+b \sim N(a\mu + b,\, a^2\sigma^2)$
3. **Sum of Gaussians:** If $X \sim N(\mu_1,\, \sigma_1^2)$ and $Y \sim N(\mu_2,\, \sigma_2^2)$ are independent, then $X + Y \sim N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$
4. **Fully characterized** by mean and variance (all higher moments determined)

When to Use

✔ Thermal noise in circuits
✔ Measurement errors
✔ Any sum of many independent random effects (CLT)
✔ Manufacturing variations
✔ When data histogram is bell-shaped and symmetric

When NOT to Use

✘ Strictly positive quantities (Gaussian allows negative values)
✘ Heavily skewed data
✘ Bounded quantities (Gaussian has infinite support)
✘ Discrete counts
✘ Heavy-tailed phenomena (use t-distribution or Cauchy)

3.10 Gamma Distribution

Physical Interpretation

Models the **total waiting time until the** $k-$ th **event** in a Poisson process, or any sum of k independent exponential random variables.

Engineering Context

- Total repair time for k components (each with exponential repair time)
- Time until $k-$ th packet arrival
- Aggregate service time for k queued jobs
- Rainfall amount modeling

Mathematical Definition

$X \sim$ Gamma$(\alpha,\ \beta)$ where α = shape parameter, β = scale parameter (or rate $\lambda = \frac{1}{\beta}$).

PDF

$$f_X(x) = \frac{x^{\alpha-1}e^{-x/\beta}}{\beta^{\alpha}\Gamma(\alpha)}, \quad x \geq 0$$

where $\Gamma(\alpha) = (\alpha-1)!$ for integer α, and $\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1}e^{-t}dt$ in general.

CDF

No closed form for general α. Available via incomplete gamma function.

Mean

$$E[X] = \alpha\beta$$

Variance

$$\text{Var}(X) = \alpha\beta^2$$

Special Cases

Parameters	Distribution
$\alpha = 1$	Exponential(β)
$\alpha = \frac{n}{2},\ \beta = 2$	Chi-square(n)
$\alpha =$ integer k	Erlang$(k,\ \beta)$

When to Use

✔ Sum of independent exponential lifetimes
✔ Waiting time for $k-$ th Poisson event
✔ Skewed, non-negative data
✔ Flexible shape (adjustable skewness)

When NOT to Use

✘ Data can be negative
✘ Data is symmetric (use Gaussian)
✘ Simpler distribution fits (exponential for $k = 1$)

3.11 Rayleigh Distribution

Physical Interpretation

Models the **magnitude (envelope)** of a $2D$ Gaussian random vector. When X and Y are independent $N(0,\, \sigma^2)$, then $R = \sqrt{X^2 + Y^2}$ follows a Rayleigh distribution.

Engineering Context

- Envelope of narrowband noise (in-phase + quadrature)
- Magnitude of wireless fading channel coefficient (Rayleigh fading)
- Wind speed modeling
- Distance error in $2D$ positioning (when errors are Gaussian in x and y)

Mathematical Definition

$R \sim$ Rayleigh(σ), where σ is the parameter (NOT the standard deviation of R).

PDF

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0$$

CDF

$$F_R(r) = 1 - e^{-r^2/(2\sigma^2)}, \quad r \geq 0$$

Mean

$$E[R] = \sigma\sqrt{\frac{\pi}{2}} \approx 1.2533\sigma$$

Variance

$$\text{Var}(R) = \frac{4-\pi}{2}\sigma^2 \approx 0.4292\sigma^2$$

Relationship to Other Distributions

- $R^2 \sim$ Exponential$(2\sigma^2)$ — the power is exponentially distributed
- If $X,\, Y \sim N(0,\, \sigma^2)$ independent, then $\sqrt{X^2 + Y^2} \sim$ Rayleigh(σ)

When to Use

✔ Envelope/magnitude of complex Gaussian signal
✔ Rayleigh fading channel modeling (no line-of-sight)
✔ $2D$ distance with Gaussian errors in each dimension
✔ RSS (root-sum-square) of two independent Gaussian components

When NOT to Use

✘ Strong line-of-sight component present (use Rician)
✘ Not modeling a magnitude/envelope
✘ Components are not Gaussian or not equal-variance

3.12 Chi-Square Distribution

Physical Interpretation

Models the **sum of squares of independent standard normal** random variables. Arises naturally in variance estimation and goodness-of-fit testing.

Engineering Context

- Sample variance distribution (for Gaussian data)
- Power of Gaussian noise (sum of squared components)
- Goodness-of-fit test statistic
- Confidence intervals for variance
- Energy detection in spectrum sensing

Mathematical Definition

If $Z_1, Z_2, \ldots, Z_n$ are independent $N(0, 1)$, then:

$$\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_n^2 \sim \chi^2(n)$$

n = degrees of freedom.

PDF

$$f_X(x) = \frac{x^{n/2-1} e^{-x/2}}{2^{n/2}\Gamma(n/2)}, \quad x \geq 0$$

CDF

No closed form for general n. Computed via incomplete gamma function.

Mean

$$E[X] = n$$

Variance

$$\text{Var}(X) = 2n$$

Relationship to Other Distributions

- Chi-square(n) = Gamma $\left(\frac{n}{2}, 2\right)$
- Chi-square(1) = square of $N(0, 1)$
- Chi-square(2) = Exponential $\left(\frac{1}{2}\right)$
- For large n: $\chi^2(n) \approx N(n, 2n)$ by CLT

When to Use

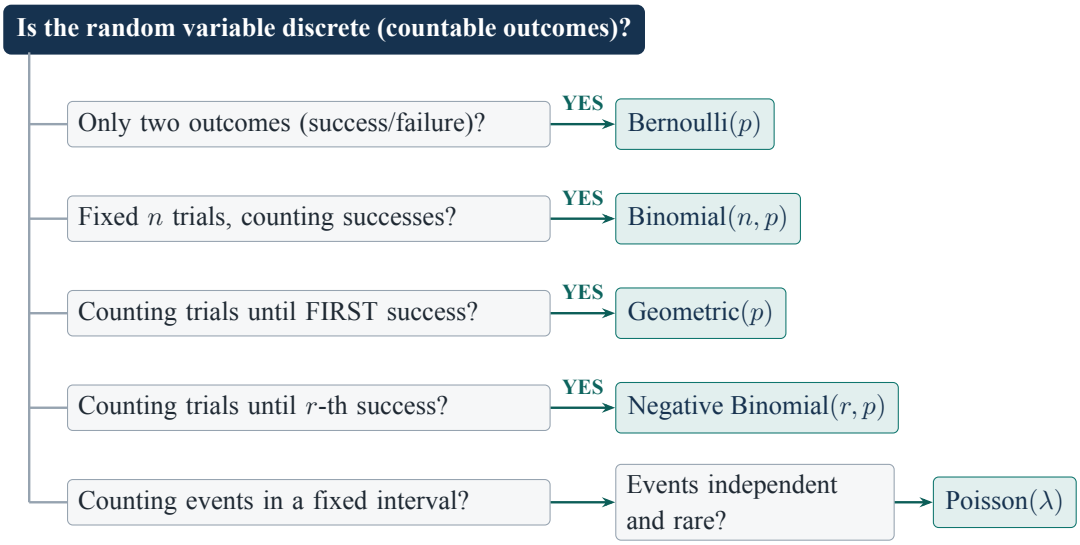
✔ Sum of squared Gaussian random variables
✔ Variance testing and confidence intervals
✔ Goodness-of-fit tests
✔ Energy detection in N dimensions

When NOT to Use

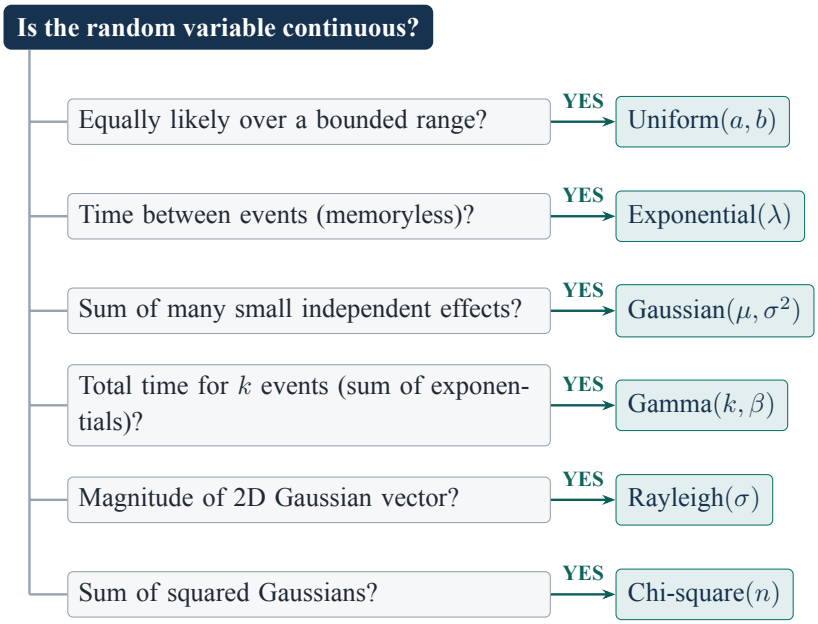
✘ Variables are not Gaussian
✘ Variables are not independent
✘ Variables are not standard normal (need scaling)

3.13 Distribution Selection Guide

Decision Tree: Discrete Distributions



Decision Tree: Continuous Distributions



Quick Recognition Patterns

If you see…	Think…
Binary outcome	Bernoulli
"n trials, k successes"	Binomial

If you see…	Think…
"How many until first…"	Geometric
"Events per interval"	Poisson
"Equal chance anywhere in $[a, b]$"	Uniform
"Time until next event"	Exponential
"Sum of many effects" or "noise"	Gaussian
"Total waiting time for k events"	Gamma
"Signal envelope" or "fading"	Rayleigh
"Sum of squared normals"	Chi-square

3.14 Summary Table

Distribution	Type	Parameters	Mean	Variance
Bernoulli	Discrete	p	p	$p(1-p)$
Binomial	Discrete	n, p	np	$np(1-p)$
Geometric	Discrete	p	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Neg. Binomial	Discrete	r, p	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Poisson	Discrete	λ	λ	λ
Uniform	Continuous	a, b	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential	Continuous	λ (or $\beta = \frac{1}{\lambda}$)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gaussian	Continuous	μ, σ^2	μ	σ^2
Gamma	Continuous	α, β	$\alpha\beta$	$\alpha\beta^2$
Rayleigh	Continuous	σ	$\sigma\sqrt{\frac{\pi}{2}}$	$\frac{(4-\pi)\sigma^2}{2}$
Chi-square	Continuous	n	n	$2n$

3.15 Practice Problems

Problem 1: Distribution Identification

Identify the appropriate distribution for each scenario:

(a) Number of dropped calls in a 10-minute window (average: 2 per 10 min)
(b) Voltage of thermal noise across a resistor
(c) Time until a server crashes (constant failure rate)
(d) Number of defective ICs out of a batch of 100 (defect rate 3%)
(e) Envelope of received signal through multipath with no LOS

Solution (a) Poisson($\lambda = 2$), (b) Gaussian(0, σ^2), (c) Exponential(λ), (d) Binomial(100, 0.03), (e) Rayleigh(σ)

Problem 2: Exponential vs Poisson

Packets arrive at a router as a Poisson process with rate $\lambda = 5$ packets/second.

(a) What is $P(\text{more than 8 packets in 1 second})$?
(b) What is $P(\text{no packet arrives in the next 0.5 seconds})$?
(c) What distribution describes the inter-arrival time?
(d) What is the mean inter-arrival time?

Solution

(a) $X \sim \text{Poisson}(5)$: $P(X > 8) = 1 - \texttt{poisscdf}(8, 5) = 0.0681$
(b) $T \sim \text{Exp}(5)$: $P(T > 0.5) = e^{-5\times 0.5} = e^{-2.5} = 0.0821$
(c) Exponential with rate $\lambda = 5$
(d) Mean $= \frac{1}{\lambda} = 0.2$ seconds

Problem 3: Gaussian Probability

A resistor has nominal value 1000Ω with manufacturing variation modeled as $N(1000,\, 25)(\sigma = 5\,\Omega)$.

(a) What fraction of resistors are within $\pm 10\,\Omega$ of nominal (spec: $990 - 1010\,\Omega$)?
(b) What fraction fail the $\pm 1\%$ tolerance test?
(c) What tolerance range contains 99.7% of resistors?

Solution

(a) $P(990 \le X \le 1010) = P(-2 \le Z \le 2) = 0.9545 \to 95.45\%$
(b) $\pm 1\% = \pm 10\,\Omega \to$ same as (a), fail rate $= 1 - 0.9545 = 4.55\%$
(c) $\pm 3\sigma = \pm 15\,\Omega \to$ range $[985,\, 1015]\Omega$

Next Module: Module 4 — Expectation and Moments